

BROWNIAN POTENTIAL THEORY

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ABSTRACT. This project surveys the relationship between Brownian hitting probabilities and capacity. We review transient Brownian motion and Green functions, then present a comparability theorem between hitting probability and Martin capacity. We conclude with an application to boundary regularity for the Dirichlet problem.

1. INTRODUCTION

Question 1.1. Let $A \in \mathcal{B}(\mathbb{R}^d)$, what is the probability that a Brownian path starting from $x \in \mathbb{R}^d$ hits A ?

$$\mathbb{P}_x(B(t) \in A, t > 0)$$

Lebesgue measure is not comparable with Brownian hitting probability. For example, in $d = 3$, Brownian motion starting at 0 almost surely hits the unit sphere, but almost surely does not hit any prescribed point on the sphere. Both sets have Lebesgue measure 0, but hitting probabilities are very distinguished from one another. This suggests the need for a different notion of size to quantify how visible a set is to Brownian motion.

The motivation comes from the physical notion of capacitance. In electrostatics, one asks how much energy is required to place a unit amount of charge on a conductor. Mathematically, this corresponds to placing a unit charge distribution, or probability measure μ , on a set A . The energy measures the average pairwise interaction of this charge distribution. The *capacity* of the set is then defined as the reciprocal of the minimal possible energy.

Large capacity means that one can place a unit amount of mass on A so that the pairwise interaction energy is small. In this sense, A is not concentrated into a tiny target: it has enough spread or thickness to support a low-energy measure. Theorem 3.1 says that this potential-theoretic thickness is precisely what Brownian motion sees.

In Section 2, we introduce the background of transient Brownian motion $\{B(t) : 0 \leq t \leq T\}$ and the Green function $G(x, y)$ as the occupation time density of $B(t)$. We then define energy and capacity. The Martin kernel is a normalized Green function normalized relative, and in Section 3 we show that the Martin capacity of A is comparable to the hitting probability of A .

Finally, in Section 4, we discuss an application to the Dirichlet problem. Let $U \subset \mathbb{R}^d$ be a bounded domain and let $\varphi : \partial U \rightarrow \mathbb{R}$ be continuous. The Brownian exit distribution formula

$$(1) \quad u(x) = \mathbb{E}_x[\varphi(B(\tau))], \quad \tau = \inf\{t > 0 : B(t) \in \partial U\}$$

gives a natural candidate solution to the Dirichlet problem

$$(2) \quad \begin{cases} \Delta u = 0 & \text{in } U \\ u|_{\partial U} = \varphi \end{cases}$$

The difficulty does not lie in the harmonicity of (1) in the interior, but rather in whether the formula continuously extends to correct boundary values. A boundary point $z \in \partial U$ is called

regular if Brownian motion started at z hits U^c immediately:

$$\mathbb{P}_z(\tau_{U^c} = 0) = 1$$

Theorem 4.2 says (1) solves (2) if every point on the boundary is regular. Thus boundary regularity is again a question of Brownian visibility. Wiener's test makes this precise by characterizing regularity in terms of the capacity of the complement near the boundary point.

2. GREEN FUNCTIONS, ENERGY, AND CAPACITY

Throughout the rest of the article, we write $\mathbb{P}_x$ and $\mathbb{E}_x$ for probability and expectation when Brownian motion starts at x .

We pose a related question to 1.1:

Question 2.1. Let $U \subset \mathbb{R}^d$ be open. What is the expected total occupation time of $B(t)$ in U ?

Recall that for a d -dimensional Brownian motion $B(t)$, the transition density is

$$(3) \quad p(t, x, y) = \frac{1}{(2\pi t)^{d/2}} \exp\left(-\frac{|x - y|^2}{2t}\right)$$

and $p(t, x, y) dy$ is the probability that Brownian motion starting at x is near y at time t . The Green function $G(x, y)$ is the spatial density of the expected occupation measure, meaning it measures the expected total time Brownian motion starting at x spends near y .

A first attempt would be to define

$$(4) \quad G(x, y) = \int_0^\infty p(t, x, y) dt$$

However, due to neighborhood recurrence of Brownian motion when $d = 2$, the above integral blows up. More precisely, when $d = 2$ and $x \neq y$,

$$p(t, x, y) = \frac{1}{2\pi t} \exp\left(-\frac{|x - y|^2}{2t}\right) \sim \frac{1}{2\pi t} \quad \text{as } t \rightarrow \infty$$

Hence $G(x, y)$ is not well-defined in $d = 2$ under definition 4. To fix this, we work with Brownian motion with a lifetime T . This motivates the following definition.

Definition 2.2. A d -dimensional Brownian motion $\{B(t) : 0 \leq t \leq T\}$ is *transient* if one of the following holds:

- (1) $d \geq 3$ and $T = \infty$;
- (2) $d \geq 2$ and T is an independent exponential time with parameter $\lambda > 0$;
- (3) $d \geq 2$ and T is the first exit time from a bounded domain D .

The domain convention is $D = \mathbb{R}^d$ in cases (1), (2).

This is a slightly broader usage of the word transient: besides free Brownian motion in $d \geq 3$, it also includes Brownian motions with lifetimes chosen so that the Green function is useful and the Markov restart property is preserved.

By [MP10, Theorem 3.30], transient Brownian motion has a transition subdensity p^*

$$\mathbb{P}_x\{B(t) \in A \text{ and } t \leq T\} = \int_A p^*(t, x, y) dy \quad \text{for every } A \subset \mathbb{R}^d \text{ Borel}$$

Definition 2.3. The *Green kernel* $G : \mathbb{R}^d \times \mathbb{R}^d \rightarrow [0, \infty]$ of a transient Brownian motion $\{B(t) : 0 \leq t \leq T\}$ is defined by

$$G(x, y) = \int_0^\infty p^*(t, x, y) dt$$

In the domain-killed case, G is considered on $D \times D$.

The following formula identifies $G(x, y)$ as the expected occupation density, turning expected occupation times into integrals against the Green kernel.

Proposition 2.4 (Occupation Formula). Let $U \subset \mathbb{R}^d$ be an open set. Then

$$\mathbb{E}_x \left[\int_0^T \mathbf{1}_{\{B(t) \in U\}} dt \right] = \int_U G(x, y) dy$$

Proof. By Tonelli's theorem,

$$\begin{aligned} \mathbb{E}_x \left[\int_0^T \mathbf{1}_{\{B(t) \in U\}} dt \right] &= \int_0^\infty \mathbb{E}_x [\mathbf{1}_{\{B(t) \in U\}} \mathbf{1}_{\{t \leq T\}}] dt = \int_0^\infty \mathbb{P}_x \{B(t) \in U, t \leq T\} dt \\ &= \int_0^\infty \int_U p^*(t, x, y) dy dt = \int_U \left(\int_0^\infty p^*(t, x, y) dt \right) dy = \int_U G(x, y) dy \end{aligned}$$

In the next step, we use this kernel to define potentials and energies of measures. Capacity will then measure how efficiently a set can support a low-energy measure.

Definition 2.5. Suppose $A \subset \mathbb{R}^d$ is a Borel set and the *kernel* $K : \mathbb{R}^d \times \mathbb{R}^d \rightarrow [0, \infty]$ is a non-negative measurable function. Then the K -energy of a measure μ on A is defined to be

$$I_K(\mu) = \iint K(x, y) d\mu(x) d\mu(y)$$

and the K -capacity of A is defined as

$$\text{Cap}_K(A) = [\inf \{I_K(\mu) : \mu \text{ a probability measure on } A\}]^{-1}$$

Since the Green function measures expected occupation, it is natural to use G , or a normalized version of G , as the interaction kernel in the energy. The energy then measures how strongly a measure supported on A interacts with itself from the perspective of Brownian occupation.

One can compute as in [MP10, Theorem 3.33] that for free Brownian motion in $d \geq 3$,

$$G(x, y) = c(d)|x - y|^{2-d}$$

This is the Newtonian kernel, which describes a power-law interaction.

Potential theory	Brownian motion
$G(x, y)$	expected occupation density near y
$I_K(\mu)$	self-interaction energy of a charge distribution
$\text{Cap}(A)$	effective Brownian target size
$\text{Cap}(A) = 0$	Brownian invisibility / polar set

Example 2.6 ($d \geq 3$). A point has zero Green capacity: the only probability measure on $\{a\}$ is δ_a , and

$$I_G(\delta_a) = G(a, a) = \infty, \quad \text{Cap}_G(\{a\}) = 0$$

Moreover, since $G(rx, ry) = r^{2-d}G(x, y)$, the energy scales by r^{2-d} , so for $r > 0$,

$$\text{Cap}_G(rA) = r^{d-2} \text{Cap}_G(A)$$

It's important to remark that this scaling law depends on the choice of a Newtonian kernel G .

3. HITTING PROBABILITY AND MARTIN CAPACITY

Theorem 3.1. Let $\{B(t) : 0 \leq t \leq T\}$ be a transient Brownian motion and let $A \subset D$ be compact. Then

$$\frac{1}{2} \text{Cap}_M(A) \leq \mathbb{P}_0(\exists 0 < t \leq T : B(t) \in A) \leq \text{Cap}_M(A)$$

It is tempting to compare hitting probability with Green capacity,

$$\text{Cap}_G(A) = \mathbb{P}_0(\exists 0 < t \leq T : B(t) \in A)$$

but this has the wrong scaling for free Brownian motion in $d \geq 3$. By Brownian scaling,

$$\mathbb{P}_0(\exists t > 0 : B(t) \in rA) = \mathbb{P}_0(\exists t > 0 : B(t) \in A)$$

whereas by Example 2.6,

$$\text{Cap}_G(rA) = r^{d-2} \text{Cap}_G(A)$$

The right kernel is instead the Martin kernel

$$M(x, y) = \frac{G(x, y)}{G(0, y)}$$

which is the Green function normalized relative to the starting point 0. In free space, $M(rx, ry) = M(x, y)$, so Martin capacity has the same dilation-invariance as the hitting probability.

Proof. Upper bound. Let

$$\tau = \inf\{0 < t \leq T : B(t) \in A\}$$

and define the hitting distribution on A by

$$\mu(E) = \mathbb{P}_0(B(\tau) \in E, \tau \leq T), \quad E \subset A$$

Thus $\mu(A) = \mathbb{P}_0(\tau \leq T)$. The idea is to use this actual hitting distribution as a test measure for capacity. If Brownian motion first hits A at x , then by the strong Markov property the future occupation near y is measured by $G(x, y)$. Since this future occupation is part of the total occupation from 0, the Martin potential of μ should be bounded by 1.

Precisely, comparing total occupation near y with occupation after first hitting A , then applying the strong Markov property and the Green occupation formula, gives

$$\int_{B(y, \epsilon)} G(0, z) dz \geq \int_A \int_{B(y, \epsilon)} G(x, z) dz d\mu(x)$$

Dividing by $\text{vol}(B(y, \epsilon))$ and letting $\epsilon \rightarrow 0$, using continuity of $G(0, \cdot)$ away from the pole and lower semicontinuity/Fatou for the potential on the right, gives

$$G(0, y) \geq \int_A G(x, y) d\mu(x)$$

Hence

$$1 \geq \int_A M(x, y) d\mu(x)$$

Integrating this inequality over $y \in A$ with respect to $d\mu(y)$, we get

$$I_M(\mu) = \int_A \int_A M(x, y) d\mu(x) d\mu(y) \leq \mu(A)$$

If $\mu(A) = 0$, the upper bound is trivial. Otherwise set $\hat{\mu} = \frac{\mu}{\mu(A)}$. Then

$$I_M(\hat{\mu}) = \frac{I_M(\mu)}{\mu(A)^2} \leq \frac{1}{\mu(A)} \implies \mu(A) \leq \frac{1}{I_M(\hat{\mu})} \leq \text{Cap}_M(A)$$

Since $\mu(A) = \mathbb{P}_0(\tau \leq T)$, this proves

$$\mathbb{P}_0(\tau \leq T) \leq \text{Cap}_M(A)$$

Lower bound. It suffices to prove that for every probability measure ν on A with finite M -energy,

$$\mathbb{P}_0(\tau \leq T) \geq \frac{1}{2I_M(\nu)}$$

and then optimize over ν . Define

$$h_\epsilon(x, y) = \int_{B(y, \epsilon)} G(x, z) dz, \quad h_\epsilon^*(x, y) = \sup_{|x-z| < \epsilon} h_\epsilon(z, y)$$

The proof uses the second moment of the normalized occupation variable

$$Z_\epsilon = \int_A \frac{1}{h_\epsilon(0, y)} \int_0^T 1_{\{B(t) \in B(y, \epsilon)\}} dt d\nu(y)$$

It measures the ν -average occupation of ϵ -balls around A , normalized so that $\mathbb{E}_0[Z_\epsilon] = 1$. Thus by Paley-Zygmund inequality

$$\mathbb{P}_0(Z_\epsilon > 0) \geq \frac{1}{\mathbb{E}_0[Z_\epsilon^2]}$$

If $Z_\epsilon > 0$, then Brownian motion hits the ϵ -neighborhood of A . Let

$$E_\epsilon = \{\exists 0 < t \leq T : \text{dist}(B(t), A) < \epsilon\}$$

Then $\{Z_\epsilon > 0\} \subseteq E_\epsilon$. Since A is compact, continuity of Brownian paths gives $E_\epsilon \downarrow \{\tau \leq T\}$ as $\epsilon \downarrow 0$; in the case $T = \infty$, transience prevents the approximating hitting times from escaping to infinity. Hence

$$\mathbb{P}_0(\tau \leq T) \geq \liminf_{\epsilon \downarrow 0} \mathbb{P}_0(Z_\epsilon > 0) \geq \liminf_{\epsilon \downarrow 0} \frac{1}{\mathbb{E}_0[Z_\epsilon^2]} \stackrel{\text{WTS}}{\geq} \frac{1}{2I_M(\nu)}$$

It remains to estimate the second moment, which is the desired **red** inequality above. Splitting $[0, T]^2$ into $s \leq t$ and $t \leq s$,

$$\mathbb{E}_0 Z_\epsilon^2 = 2\mathbb{E}_0 \int_0^T ds \int_s^T dt \iint \frac{1_{\{B(s) \in B(x, \epsilon), B(t) \in B(y, \epsilon)\}}}{h_\epsilon(0, x)h_\epsilon(0, y)} d\nu(x)d\nu(y)$$

For fixed s, x, y , the strong Markov property at time s gives, on the event $B(s) \in B(x, \epsilon)$,

$$\mathbb{E}_0 \left[\int_s^T 1_{\{B(t) \in B(y, \epsilon)\}} dt \middle| \mathcal{F}_s \right] = h_\epsilon(B(s), y) \leq h_\epsilon^*(x, y)$$

Therefore

$$\mathbb{E}_0 Z_\epsilon^2 \leq 2 \iint \frac{h_\epsilon^*(x, y)}{h_\epsilon(0, x)h_\epsilon(0, y)} \mathbb{E}_0 \left[\int_0^T 1_{\{B(s) \in B(x, \epsilon)\}} ds \right] d\nu(x)d\nu(y)$$

Using the definition of $h_\epsilon(0, x)$,

$$\mathbb{E}_0 Z_\epsilon^2 \leq 2 \iint \frac{h_\epsilon^*(x, y)}{h_\epsilon(0, y)} d\nu(x) d\nu(y)$$

The remaining input is the Green-kernel estimate from [MP10, Proposition 8.23]:

$$\frac{h_\epsilon^*(x, y)}{h_\epsilon(0, y)} \leq CM(x, y), \quad \frac{h_\epsilon^*(x, y)}{h_\epsilon(0, y)} \rightarrow M(x, y)$$

Since $I_M(\nu) < \infty$, dominated convergence gives

$$\limsup_{\epsilon \downarrow 0} \mathbb{E}_0 Z_\epsilon^2 \leq 2 \iint M(x, y) d\nu(x) d\nu(y) = 2I_M(\nu)$$

Combining with Paley–Zygmund,

$$\mathbb{P}_0(\tau \leq T) \geq \frac{1}{2I_M(\nu)}$$

Taking the supremum over probability measures ν on A ,

$$\mathbb{P}_0(\tau \leq T) \geq \frac{1}{2} \sup_{\nu \in \mathcal{P}(A)} \frac{1}{I_M(\nu)} = \frac{1}{2} \text{Cap}_M(A)$$

This proves the lower bound. □

4. AN OUTLOOK: THE DIRICHLET PROBLEM

We give an application of Theorem 3.1 to the Dirichlet problem.

Question 4.1. Let $U \subset \mathbb{R}^d$ be a bounded domain and let $\varphi : \partial U \rightarrow \mathbb{R}$ be continuous. When is the Brownian exit distribution formula

$$u(x) = \mathbb{E}_x[\varphi(B(\tau_U))], \quad \tau_U = \inf\{t > 0 : B(t) \notin U\}$$

the solution to

$$\begin{cases} \Delta u = 0 & \text{in } U \\ u|_{\partial U} = \varphi \end{cases}$$

The interior harmonicity is not the difficult part. If $B(x, r) \Subset U$ and $\tau_r = \inf\{t > 0 : B(t) \in \partial B(x, r)\}$, then the strong Markov property gives

$$u(x) = \mathbb{E}_x[\varphi(B(\tau_U))] = \mathbb{E}_x[\mathbb{E}_{B(\tau_r)}[\varphi(B(\tau_U))]] = \mathbb{E}_x[u(B(\tau_r))]$$

Since Brownian motion exits a ball uniformly on the boundary sphere,

$$u(x) = \frac{1}{|\partial B(x, r)|} \int_{\partial B(x, r)} u(y) d\sigma(y)$$

Thus, assuming the Brownian formula is locally bounded, u is harmonic in the interior by [MP10, Theorem 3.8].

The real difficulty is the boundary behavior. Formally, if $z \in \partial U$ and Brownian motion exits immediately, then $u(z) = \mathbb{E}_z[\varphi(B(0))] = \varphi(z)$. But the Dirichlet problem asks for the interior function to extend continuously to the boundary:

$$\lim_{\substack{x \in U \\ x \rightarrow z}} \mathbb{E}_x[\varphi(B(\tau_U))] = \varphi(z), \quad z \in \partial U$$

Intuitively, Brownian motion started near z should exit near z . Equivalently, Brownian motion started at z should see U^c immediately. We say $z \in \partial U$ is *regular* for U^c if

$$\mathbb{P}_z(\tau_{U^c} = 0) = 1, \quad \tau_{U^c} = \inf\{t > 0 : B(t) \in U^c\}$$

Theorem 4.2. Let $U \subset \mathbb{R}^d$ be a bounded domain and let $\varphi : \partial U \rightarrow \mathbb{R}$ be continuous. If every $z \in \partial U$ is regular for U^c , then

$$u(x) = \mathbb{E}_x[\varphi(B(\tau_U))], \quad \tau_U = \inf\{t > 0 : B(t) \notin U\}$$

solves the Dirichlet problem in U with boundary values φ .

Instead of proving Theorem 4.2, we give a short example where failure to see the complement immediately causes trouble.

Example 4.3 ($d = 2$). Let $D = \mathcal{B}(0, 1) \subset \mathbb{R}^2$, and let $u : D \rightarrow \mathbb{R}$ solve the Dirichlet problem with boundary data $\varphi : \partial D \rightarrow \mathbb{R}$. By the Brownian representation formula for the disk [MP10, Theorem 3.12],

$$u(x) = \mathbb{E}_x[\varphi(B(\tau_D))], \quad \tau_D = \inf\{t > 0 : B(t) \notin D\}$$

Now consider the punctured disk $D \setminus \{0\}$, whose boundary is $\partial D \cup \{0\}$. Keep the same values on ∂D , and prescribe $\varphi(0)$ at the puncture. If $\varphi(0) \neq u(0)$, then the Brownian formula does not solve the Dirichlet problem on $D \setminus \{0\}$.

Indeed, planar Brownian motion almost surely does not hit a fixed point. Hence the exit time $\tau_{D \setminus \{0\}} = \inf\{t > 0 : B(t) \notin D \setminus \{0\}\}$ agrees almost surely with τ_D , so for $x \in D \setminus \{0\}$,

$$\mathbb{E}_x[\varphi(B(\tau_{D \setminus \{0\}}))] = \mathbb{E}_x[\varphi(B(\tau_D))] = u(x)$$

Taking $x \rightarrow 0$, the formula approaches $u(0)$, not the prescribed value $\varphi(0)$. Thus the puncture is invisible to the exit distribution.

What is the sharp condition for regularity? The answer is Wiener's test. For a closed set Λ , define

$$A_x(k) = \{y \in \mathbb{R}^d : 2^{-k-1} \leq |y - x| \leq 2^{-k}\}, \quad \Lambda_x^k = \Lambda \cap A_x(k)$$

Thus Λ_x^k is the part of Λ in the k -th dyadic shell around x .

Theorem 4.4 (Wiener's test). A point $x \in \mathbb{R}^d$ is regular for the closed set $\Lambda \subset \mathbb{R}^d$, $d \geq 3$, if and only if

$$\sum_{k=1}^{\infty} 2^{k(d-2)} C_{d-2}(\Lambda_x^k) = \infty$$

where C_{d-2} is the Newtonian capacity.

The intuition is that regularity is a scale-by-scale hitting problem. Near x , Brownian motion must see Λ immediately. Theorem 3.1 says hitting probabilities are controlled by capacity, so Wiener's test adds the capacity contributions from all small dyadic shells. If the sum diverges, Λ has enough Brownian-visible thickness near x ; if it converges, the set is too thin near x .

Example 4.5 (Lebesgue's thorn). In $\mathbb{R}^3$, let

$$G = \{(x_1, x_2, x_3) \in (-1, 1)^3 : \sqrt{x_2^2 + x_3^2} > x_1^\alpha \text{ if } x_1 \geq 0\}$$

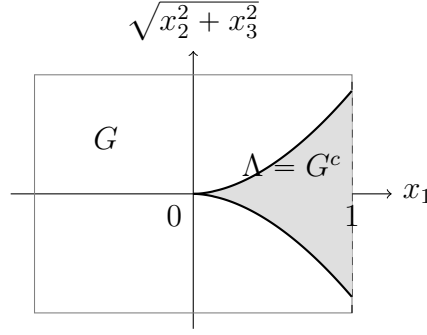


FIGURE 1. A two-dimensional cross-section of Lebesgue's thorn

Then G has a cusp at the origin, and its complement $\Lambda = G^c$ is a thin thorn along the positive x_1 -axis. When $\alpha > 1$, the thorn narrows too quickly: in the k -th annulus its radius is roughly $2^{-\alpha k}$, giving

$$\sum_{k=1}^{\infty} 2^k C_1(\Lambda_0^k) \leq \sum_{k=1}^{\infty} 2^k 2^{-\alpha k} < \infty$$

so the origin is irregular for Λ . For $\alpha \leq 1$, the complement is thick enough near the origin, and the origin is regular by the cone condition. This illustrates the point of Wiener's test: boundary regularity is governed by capacity at small scales, not smoothness alone.

5. CONCLUSIONS

In this article, we explored the notion of capacity - analogous to that of capacitance from physics - as a scale of Brownian visibility. In particular, we shown that the Martin capacity of a set A is comparable up to a factor of two to the hitting probability of a transient brownian motion onto A .

The question of quantifying hitting probability (or finding comparables) is of great importance to the Dirichlet problem. The exit distribution formula

$$u(x) = \mathbb{E}_x[\varphi(B(\tau))]$$

solves the Dirichlet problem in U with boundary φ under regularity of the boundary for the complement U^c . Wiener's test quantifies this sharp condition of u being a solution to the Dirichlet problem.

REFERENCES

[MP10] Peter Mörters and Yuval Peres, *Brownian motion*, Cambridge University Press, 2010.